Foundations of Graphics & Visual Computing

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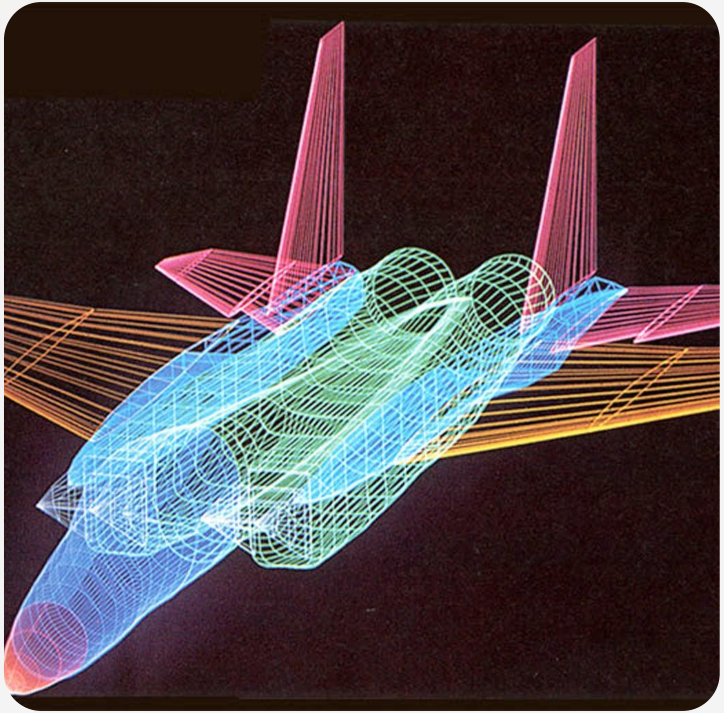
CS ELEC 3 / PROF ELEC 2

Overview of Computer Graphics and Visual Computing

Computer Graphics

- the discipline concerned with creating, representing, and manipulating visual content using computers.
- the process often involves algorithms and mathematical models to simulate real-world visuals or generate artistic designs.
- It's not just about creating pretty pictures; it’s about a pipeline of processes that transforms data into images.



Vector Graphics

- these images are defined by mathematical formulas that describe shapes, lines, and curves.
- based on formulas, they can be scaled to any size without losing quality or becoming pixelated.
- A logo is a common example.

Raster Graphics

- also known as **bitmap graphics**
- these images are made of a grid of tiny dots called pixels (picture elements), containing color information.
- photographs and digital paintings are raster images.
- when scaled up, they can appear blocky and lose clarity.

(a) Original image (b) Sharpened image (c) Blurry image

Core Concepts

Modeling

- This is the process of defining the shape and form of a 3D object. It's like sculpting a virtual model.

Animation

- The illusion of movement created by rapidly displaying a sequence of still images (frames).

Rendering

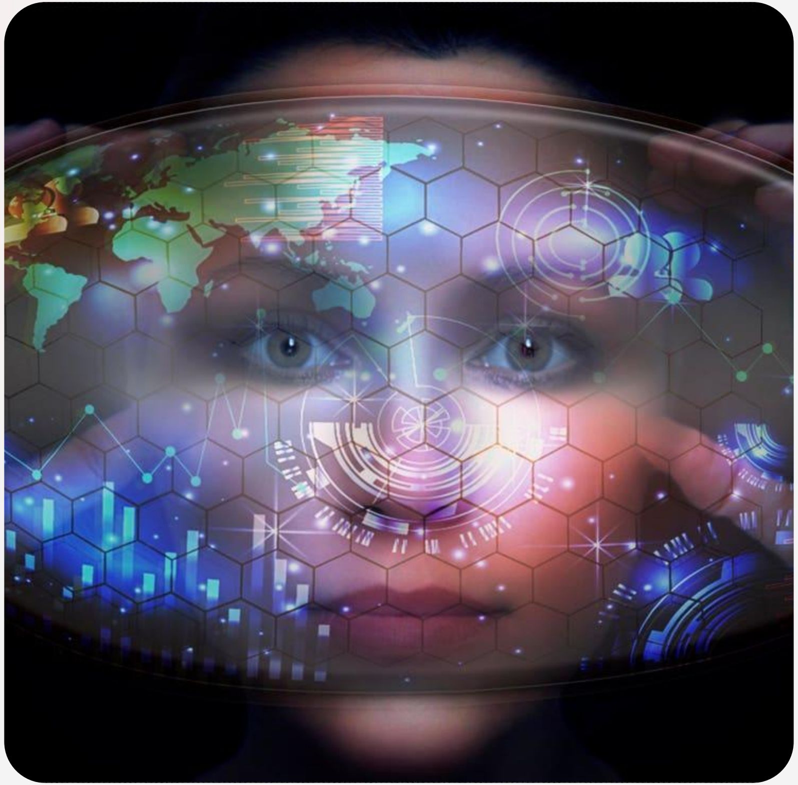
- The process of generating an image from a 3D model. This involves calculating light interaction and texture.

Visual Computing

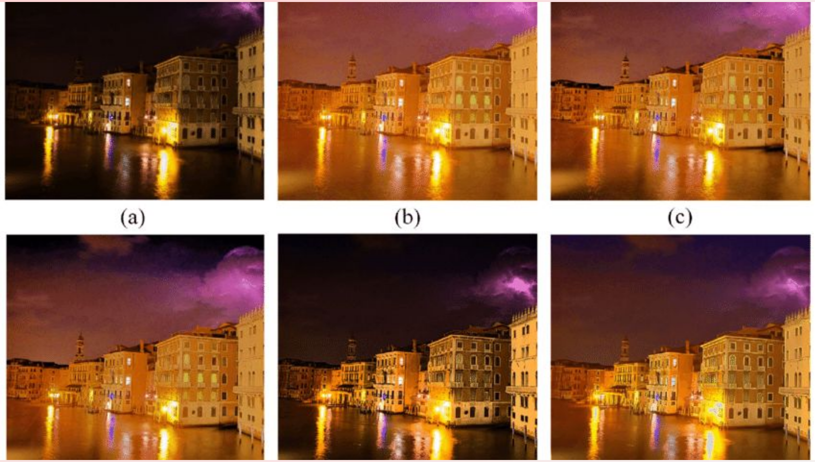
- A broader term that includes not only computer graphics but also computer vision and image processing.

Visual Computing

- a broader field encompassing computer graphics, image processing, and computer vision.
- deals with capturing, manipulating, and interpreting visual information from images and videos, empowering applications such as virtual reality (VR) and augmented reality (AR).



Key Subfields



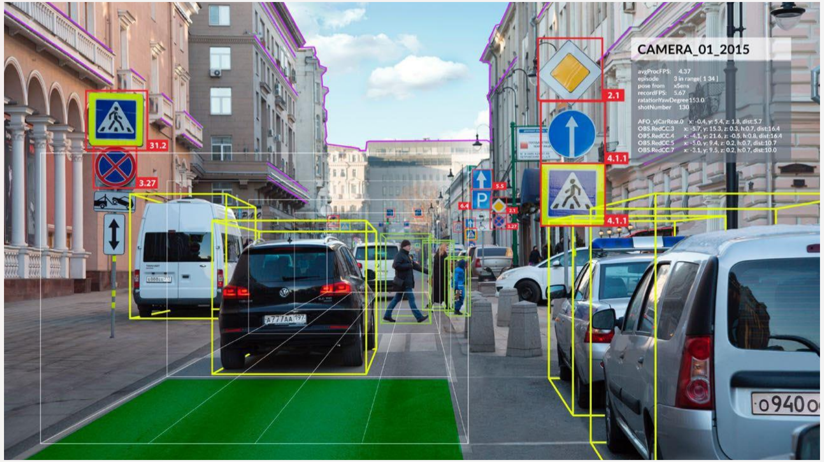


Image Processing

- Techniques for enhancing and analyzing digital images (e.g., noise reduction, segmentation).

Computer Vision

- Enabling computers to interpret and understand images and video content.

Key Subfields

Virtual Reality

- Creating immersive simulated environments where users can interact digitally.

Augmented Reality

- Overlaying digital content on the real world, enhancing user interaction with their surroundings.

History and Evolution of Computer Graphics

Early Beginnings (1950s-1960s)

- The **SAGE** (*Semi-Automatic Ground Environment*) system, an air-defense system, used the first interactive graphics display with a light pen. This was a pivotal moment, as it allowed direct interaction with a computer for the first time.

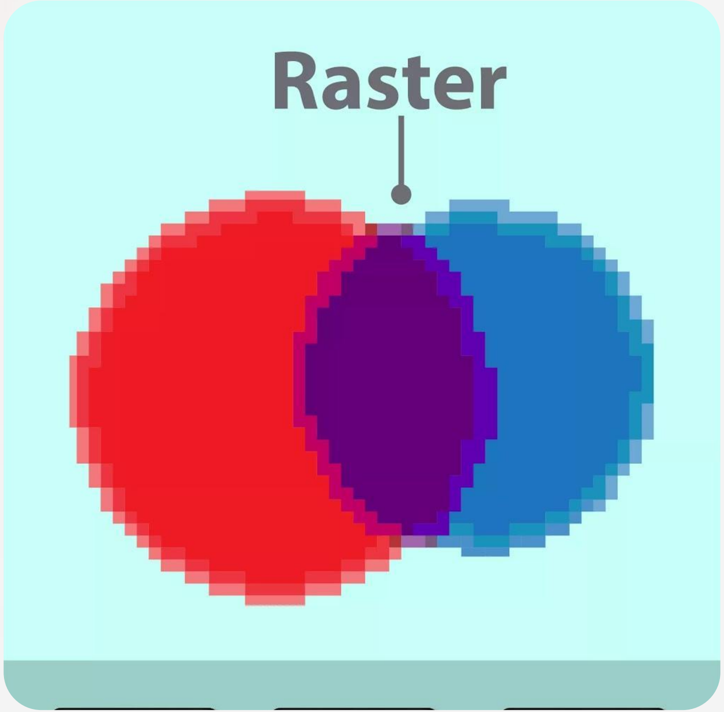
Early Beginnings (1950s-1960s)

- Initial use of **cathode ray tube** (CRT) displays to create simple line-based vector graphics.
- Development of the first computer-based drawing systems like Ivan Sutherland's **Sketchpad** (1963), which introduced interactive graphical interfaces using a light pen.

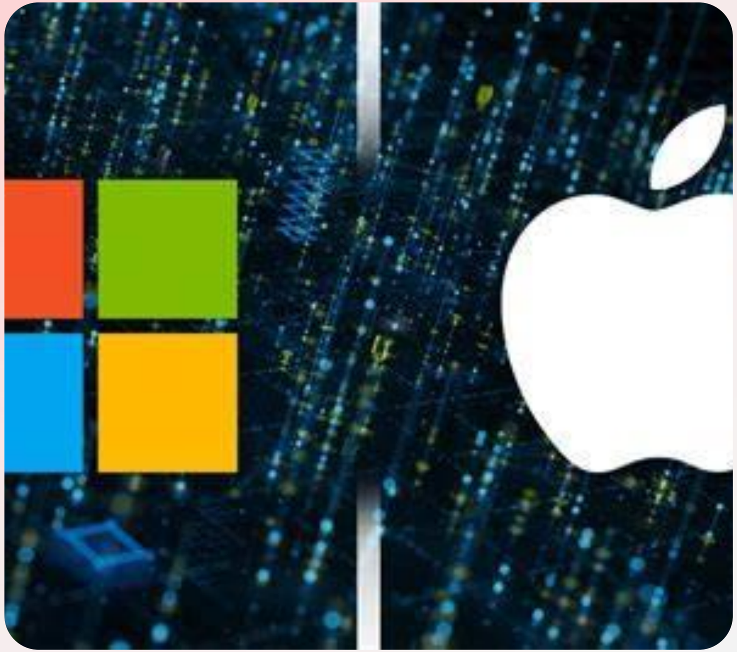
Raster Graphics and Pixels (1950s-1970s)

- Introduction of **raster graphics**, representing images as grids of pixels, which enabled more complex and realistic imagery.

- Development of **graphical user interfaces** (GUIs) at Xerox PARC in the 1970s, making computers more accessible and visually engaging.



Advances in Rendering and Animation (1980s-present)



- This decade brought graphics to the masses. Companies like Apple and Microsoft popularized **GUI**s with the Macintosh and Windows operating systems. Dedicated graphics cards (like the IBM CGA) emerged, and the first significant uses of computer graphics in film, such as in Tron (1982), began to appear.

Advances in Rendering and Animation (1980s-present)

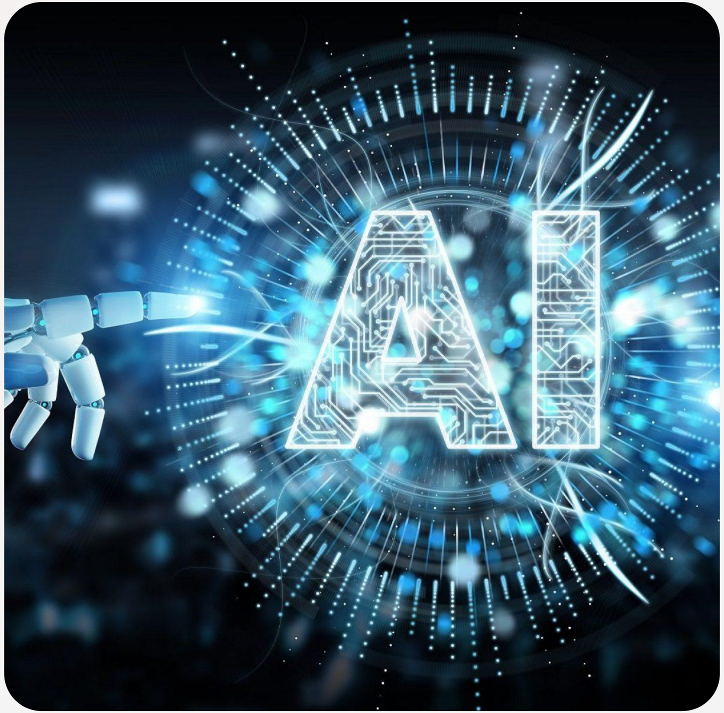
- Introduction of **shading**, **texture mapping**, and **photorealistic rendering techniques**.
- Growth of **3D graphics** fueled by the rise of gaming consoles and PC graphics accelerators like the 3dfx Voodoo card. This made real-time 3D rendering accessible to consumers.

Advances in Rendering and Animation (1980s-present)

- The focus shifted to real-time photorealism and high-fidelity rendering. Innovations like **ray tracing** (simulating light rays) became standard, and the field has diversified into new areas like **Augmented Reality** (AR) and **Virtual Reality** (VR), requiring even more advanced rendering techniques and performance.

Modern Era

- Integration of visual computing with artificial intelligence for enhanced image recognition and interpretation.
- Emergence of VR and AR technologies enhancing immersive and interactive experiences.



Applications in Modern Computing

Entertainment & Media

Video Games

- Driving much of the innovation in real-time 3D graphics, physics engines, and advanced rendering techniques.

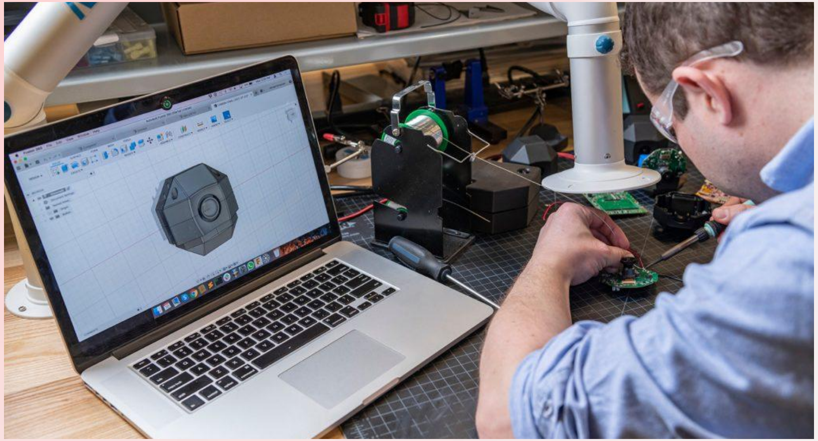
Film & Television

- Used extensively for special effects (CGI), creating animated films, and pre-visualization of scenes.

V and A Reality

- Creating immersive experiences for gaming, education, and social interaction.

Science & Engineering



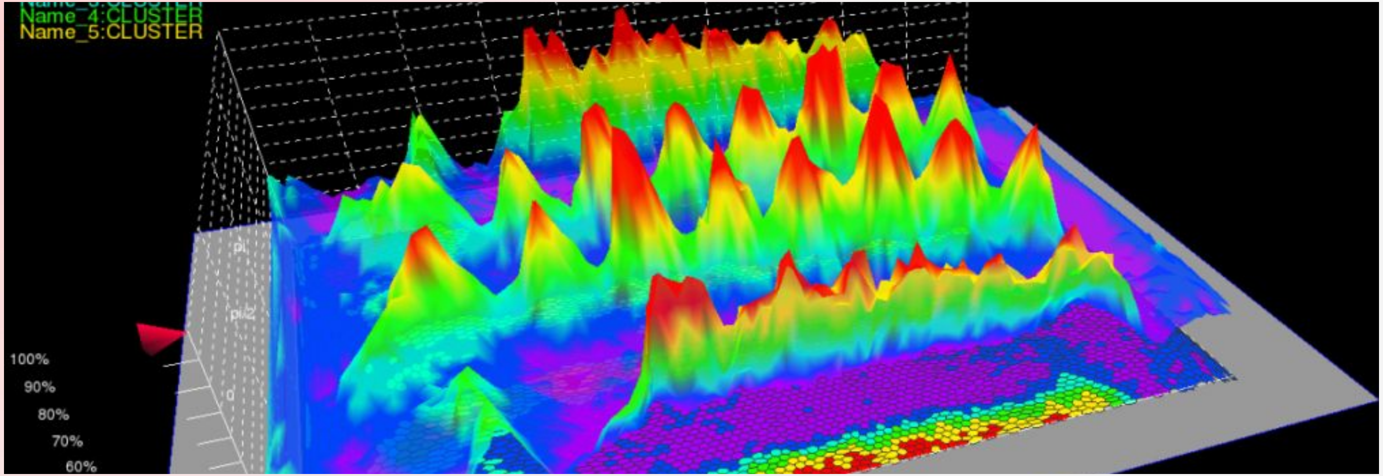
Computer-Aided Design

- Used by architects, engineers, and industrial designers to create precise digital models of products and buildings, replacing traditional blueprints.

Medical Imaging

- Doctors use visual computing to analyze and interpret medical scans like MRIs and CT scans, visualizing internal structures in 3D to aid diagnosis and surgical planning.

Science & Engineering



Scientific Visualization

- Transforming complex data sets—from climate simulations to molecular biology—into understandable visual representations to aid research and analysis.

Business & Communication

User Interfaces

- The GUIs on our smartphones, computers, and tablets are all a product of computer graphics, making technology intuitive and accessible.

Data Visualization

- Creating charts, graphs, and infographics to represent large amounts of data visually, helping businesses and analysts make informed decisions.

Business & Communication

E-commerce

- Providing 3D models and augmented reality previews of products, allowing customers to visualize items in their homes before buying.